STATA Tutorial 4B

Summary Statistics

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Outline

Summary Statistics

Generate Publishable Tables

Survey Data and Weight

- Again, recall our knowledge on descriptive statistics
- For describing a single variable, what are the proper statistics?
  - Continuous variable:
  - Discrete variable:
- For describing bivariate relationships?
  - Discrete + Continuous:
  - Continuous + Continuous:
  - Discrete + Discrete:

Summarize for Numerical Data: "summarize"

```
. summarize income

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      income |      2,488    12611.56    20656.22          1     500000

. sum loginc //sum for short

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      loginc |      2,488    8.599685    1.586994   .6931472   13.12236

. sum loginc, detail //detail for detailed statistics

                    Logged Annual Earnings
-------------------------------------------------------------
      Percentiles      Smallest
 1%     2.397895       .6931472
 5%      5.70711       .6931472
10%     6.685861       .6931472       Obs               2,488
25%     7.824446       .6931472       Sum of Wgt.       2,488

50%     8.853808                      Mean           8.599685
                        Largest       Std. Dev.      1.586994
75%     9.615872       12.20608
90%     10.30899       12.42922       Variance       2.518551
95%     10.59666       12.61154       Skewness      -1.348475
99%     11.46164       13.12236       Kurtosis         6.6859
```

Tabulate for Categorical Data: "tabulate"

```
. tabulate gender //marginal distribution
    Gender |      Freq.     Percent        Cum.
-----------+-----------------------------------
    Female |      1,770       52.68       52.68
      Male |      1,590       47.32      100.00
-----------+-----------------------------------
     Total |      3,360      100.00
. tab gender, nolab
    Gender |      Freq.     Percent        Cum.
-----------+-----------------------------------
         0 |      1,770       52.68       52.68
         1 |      1,590       47.32      100.00
-----------+-----------------------------------
     Total |      3,360      100.00
.         //tab for short; nolab to suppress value labels
. tab female
    Female |      Freq.     Percent        Cum.
-----------+-----------------------------------
         0 |      1,590       47.32       47.32
         1 |      1,770       52.68      100.00
-----------+-----------------------------------
     Total |      3,360      100.00
. summarize female
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      female |      3,360    .5267857    .4993563          0          1
.         //think about it, why the mean value equals %female
```

Subgroup Summary Statistics

```
. * Method 1
. bysort gender: sum income

-> gender = Female

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      income |      1,200    9756.157    16510.64          1     250000

-> gender = Male

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      income |      1,288    15271.87    23576.87          1     500000

. * Method 2
. tab gender, sum(loginc)

             | Summary of Logged Annual Earnings
    Gender   |        Mean   Std. Dev.       Freq.
-------------+------------------------------------
     Female  |   8.2122434   1.7203775       1,200
       Male  |   8.9606563   1.3561561       1,288
-------------+------------------------------------
      Total  |   8.5996855   1.5869944       2,488
```

Two-way Tabulate Table

```
. *help tabulate
. tab urban hukou //tabulate twoway
```

Residence (Urban/Rural)	Hukou: Rural	Non-Rural	Others	Total
Rural Areas	1,663	129	2	1,794
Urban Areas	678	886	1	1,565
Total	2,341	1,015	3	3,359

```
. tab urban hukou, cell //joint distribution
```

Key
frequency *cell percentage*

Residence (Urban/Rural)	Hukou: Rural	Non-Rural	Others	Total
Rural Areas	1,663	129	2	1,794
	49.51	3.84	0.06	53.41
Urban Areas	678	886	1	1,565
	20.18	26.38	0.03	46.59
Total	2,341	1,015	3	3,359
	69.69	30.22	0.09	100.00

Two-way Tabulate Table

```
. tab urban hukou, row //conditional distribution
```

Key
frequency *row percentage*

Residence (Urban/Rural)	Hukou: Rural	Non-Rural	Others	Total
Rural Areas	1,663	129	2	1,794
	92.70	7.19	0.11	100.00
Urban Areas	678	886	1	1,565
	43.32	56.61	0.06	100.00
Total	2,341	1,015	3	3,359
	69.69	30.22	0.09	100.00

```
. tab urban hukou, column //conditional distribution
```

Key
frequency *column percentage*

Residence (Urban/Rural)	Hukou: Rural	Non-Rural	Others	Total
Rural Areas	1,663	129	2	1,794
	71.04	12.71	66.67	53.41
Urban Areas	678	886	1	1,565
	28.96	87.29	33.33	46.59
Total	2,341	1,015	3	3,359
	100.00	100.00	100.00	100.00

Correlation Coefficient: "correlate"

```
. cor loginc eduy
(obs=2,488)
```

	loginc	eduy
loginc	1.0000	
eduy	0.4273	1.0000

```
. pwcorr loginc eduy
```

	loginc	eduy
loginc	1.0000	
eduy	0.4273	1.0000

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Table Output

- Don't use screenshots of the stata output!
- Two ways to generate publishable tables
  1. Method 1: copy table
  2. Method 2: some user-written packages for output, here I use "esttab"

Table Output

- Method 2 is recommended. We want to write down the codes to automate the output for several reasons
  1. Normally for a project, we will have many "control variables".
  2. When you work on a project, you may need to revise your tables many times.

```
. est clear
. qui bysort female: eststo: estpost sum income loginc
. qui eststo: estpost sum income loginc
.
. esttab est1 est2 est3 using "$outer/output/descrip.csv", ///
>         replace cell(mean(fmt(%10.3fc)) sd(fmt(%10.3fc)par)) ///
>         label nonumber nomtitle nogaps ///
>         title("Average Income by Gender") ///
>         note("Standard deviations in parentheses.")
(output written to /Users/bing/Dropbox/MGCS5010_2023Spring/1.Lectures/STA
> TATutorial/output/descrip.csv)
```

Table: Average Income by Gender

	Male	Female	Total
Annual Total Income	15271.870	9756.157	12611.558
	(23576.874)	(16510.641)	(20656.221)
Logged Annual Earnings	8.961	8.212	8.600
	(1.356)	(1.720)	(1.587)
Observations	1288	1200	2488

Notes: Standard deviations in parentheses.

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Survey Data and Weight

Survey Data and Weight

- CFPS uses the multi-stage probability sample method for efficiency and precision. The probability for every Chinese citizen to get selected, however, is not equal, depending on the strata.
- In surveys, almost certainly there will be non-responders.
- We assign surveyed individuals different "weights" for adjustment, and make the sample representative of the population.

Survey Data and Weight

```
. help weight

.
. sum loginc
    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
      loginc |      2,488    8.599685    1.586994   .6931472   13.12236
. sum loginc [iweight=rswt_nat]
    Variable |      Obs      Weight        Mean   Std. Dev.       Min        Max
-------------+-----------------------------------------------------------------
      loginc |    2,488  75374609.2     8.70557   1.526002   .6931472   13.12236
.
. tab gender
    Gender |      Freq.     Percent        Cum.
-----------+-----------------------------------
    Female |      1,770       52.68       52.68
      Male |      1,590       47.32      100.00
-----------+-----------------------------------
     Total |      3,360      100.00
. tab gender [iweight=rswt_nat]
    Gender |      Freq.     Percent        Cum.
-----------+-----------------------------------
    Female |  51412618.3       50.93       50.93
      Male |  49537847.5       49.07      100.00
-----------+-----------------------------------
     Total | 100,950,466      100.00
```